

Autostereoscopic Transparent Display for High-Fidelity 3D Imaging without Resolution Loss

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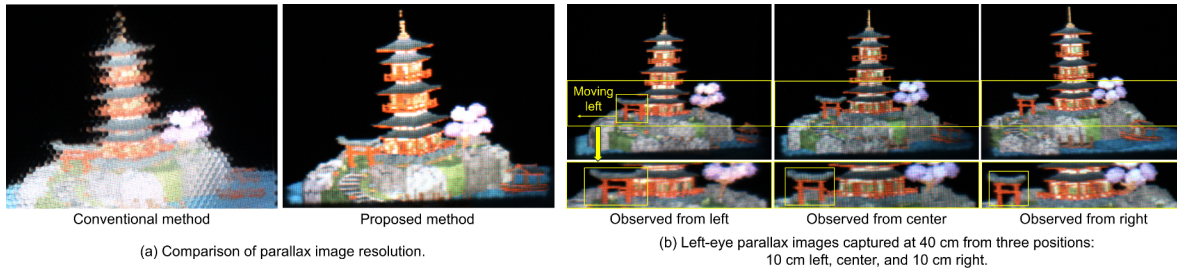


Figure 1: Captured results of parallax images

ABSTRACT

We propose a transparent display system capable of presenting high resolution autostereoscopic 3D images that naturally blend into the real-world environment. Conventional methods struggle to achieve lifelike aerial 3D representations due to resolution degradation and the conspicuous physical presence of the display hardware itself. Our proposed method combines a directional display, which presents high-resolution autostereoscopic images, with a transparent reflector to create the visual sensation of a physical 3D object floating in mid-air.

Index Terms: 3D, directional display, transparent display, autostereoscopy

1 INTRODUCTION

Driven by the long-standing demand for natural and intuitive visual experiences, numerous studies have been conducted on stereoscopic and aerial imaging technologies utilizing binocular parallax. Yet, the sense of immersion and visual fidelity in these systems are often compromised by the degradation of image resolution and the strong physical presence of the display device itself.

As a stereoscopic technology, light field displays have been proposed [1]; however, this approach suffers from resolution degradation, an issue inherent to lens array systems. Furthermore, since these displays typically present 3D images only in the immediate vicinity of the screen and do not address the visibility of the hardware itself, they fail to achieve the illusion of an image floating freely in mid-air. To realize such floating expressions, researchers have proposed methods utilizing Pepper's ghost [2] and methods that form real images in mid-air [3]. Nevertheless, the issue of resolution degradation remains unresolved in these approaches.

2 PROPOSED METHOD

2.1 Overall Overview

We propose a transparent display system designed to present autostereoscopic 3D images, with the goal of achieving high-fidelity

visuals that seamlessly integrate with the real-world environment. An overview of the system is shown in Fig. 2. The system consists of a directional display, which overcomes the resolution degradation inherent in conventional methods, and a transparent reflector that mitigates the physical presence of the hardware.

The directional display renders parallax images based on the viewpoint position. It achieves autostereoscopic viewing through time-division multiplexing, where parallax images for the left and right eyes are switched at high speed. This temporal approach prevents resolution loss in the 3D image, preserving image quality.

The light emitted from the directional display is separated into transmitted and reflected components by the transparent reflector using the Pepper's ghost technique. The system controls the light directionality by dynamically adjusting the image position on the directional display so that reflected light is directed toward the observer. This mechanism allows the observer to view the floating, autostereoscopic 3D image from arbitrary positions. Additionally, by incorporating observer tracking, the system supports simultaneous observation from multiple viewpoints, allowing multiple users to view the image at the same time.

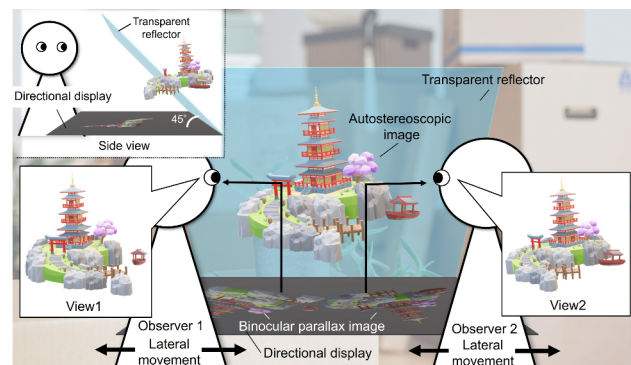


Figure 2: Overview of the proposed system

2.2 Structure of the Directional Display

The directional display consists of a light source, an optical element that modulates the image based on the observation position, and an image display unit composed of liquid crystal elements that

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display the visual content. To present the appropriate image to the observer's position, an optical element consisting of an array of plano-convex lenses is placed in front of the light source. This setup generates source light with directionality targeting the observation position. By directing this source light onto the parallax image shown on the image display unit, directionality is imparted to the parallax image.

Stereoscopic presentation using this directional display is achieved by rapidly switching directional parallax images for the observer's left and right eyes using time-division multiplexing (Fig. 3(b)). In conventional light field displays, where optical elements are placed on top of the LCD element, increasing the number of viewpoints reduces the resolution of the parallax image displayed, resulting in a degradation of the 3D image resolution (Fig. 3(a)). In contrast, our method places the optical element on the light source, thereby enabling stereoscopic presentation with a resolution equivalent to that of the image display unit.

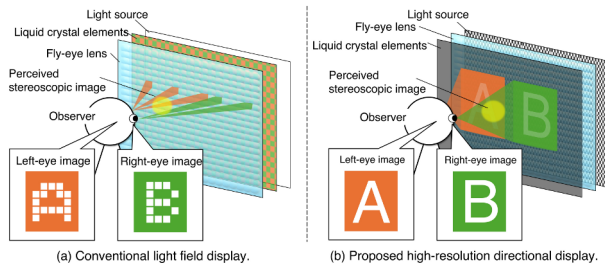


Figure 3: Comparison of stereoscopic presentation methods

To determine the directional emission for a specific observation position, our method calculates the corresponding light source pixels by simulating a ray emitted from the observer's eye. We model each lens element as a thick plano-convex lens and perform refraction calculations at both the curved surface and the flat base of the lens.

Let n_{surf} be the unit normal vector at a point P_{surf} on the lens surface, and d_{in} be the incident ray vector. Let n_{air} be the refractive index of air and n_{lens} be that of the lens material. Using the relative refractive index $\eta_1 = n_{air}/n_{lens}$, the refraction vector d_{lens} propagating into the lens is determined by Snell's law as follows:

$$d_{lens} = \eta_1 d_{in} - \left(\eta_1 (n_{surf} \cdot d_{in}) + \sqrt{1 - \eta_1^2 (1 - (n_{surf} \cdot d_{in})^2)} \right) n_{surf} \quad (1)$$

The ray then travels in a straight line through the lens interior and reaches the lens base. After calculating the intersection point P_{base} between the lens base and the ray d_{lens} , the refraction is calculated again. Let n_{base} be the normal vector of the lens base (typically in the direction of the optical axis) and $\eta_2 = n_{lens}/n_{air}$ be the relative refractive index. The vector d_{out} exiting from the lens into the air is calculated similarly using Snell's law. The intersection point of this vector d_{out} with the light source display determines the pixel to be illuminated to provide directionality toward the observer.

2.3 Realizing a Floating Virtual Image Display

As shown in Fig. 2, placing a transparent reflector above the directional display reduces the physical presence of the display device, allowing the projected image to blend seamlessly into the surrounding space and creating the illusion of an image floating in mid-air. The image presented on the directional display is reflected by the transparent plate. The directionality of the display is controlled to account for this reflection path. When the observer views the

reflected light, the Pepper's ghost effect creates the visual sensation that the stereoscopic image is suspended in the air. Furthermore, by tracking the observer, the system ensures that the appropriate stereoscopic image is observable from arbitrary positions, as demonstrated in Fig. 1(b).

3 RESULTS

To verify the parallax image directionality essential for our method, we constructed virtual and real prototypes (Fig. 4(a)). The setup comprises a fly-eye lens atop a light-source display, a backlightless LCD serving as the image display unit, and a 45-degree plane mirror acting as the transparent reflector. The simulation calculated optimal pixel illumination for directionality.

We used two cameras (BFS-U3-200S6C) equipped with markers for optical motion capture, positioned at the interpupillary distance to capture the directional images. The experimental results in Fig. 4(b) show the parallax images for the left and right eyes captured by the two cameras. The results indicate that the left-eye camera clearly captured the parallax image directed toward the left eye, while the parallax image directed toward the right eye was only faintly visible. Similar results were obtained for the right-eye camera. These results confirm that image presentation with appropriate directionality is achieved when operating the prototype based on the simulation calculations. We also confirmed that stereoscopic viewing remains feasible with this level of crosstalk.

Finally, Fig. 1(a) demonstrates that unlike the conventional method which suffers resolution degradation, the proposed method maintains high resolution.

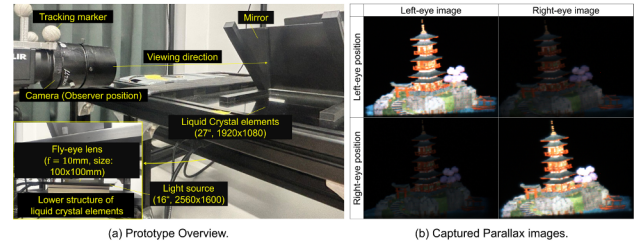


Figure 4: Prototype and experimental results

4 CONCLUSION

We proposed a high-resolution autostereoscopic transparent display. Experiments confirmed accurate directionality for arbitrary positions, demonstrating the system's feasibility. Future work includes using a transparent plate for seamless integration and extending the reflector to polyhedral or curved surfaces to expand the viewing zone.

ACKNOWLEDGMENTS

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